

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (IT)

May 2012

IT206 Object Oriented Programming using Java

Date: 18.05.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q - 1 Attempt following.

[07]

- (a) Perform match making between A and B concepts.

A

- (a) multiple inheritance
- (b) finalize
- (c) Object
- (d) protected members
- (e) static
- (f) package
- (g) interface

B

- (a) collection of final variables, abstract methods
- (b) different package and non-subclass
- (c) not supported in java
- (d) same package and subclass
- (e) collection of classes, interfaces and packages.
- (f) garbage collection
- (g) super most class
- (h) class variable
- (i) instance variable

Q - 2 Answer the following.

- (a) Create an abstract class Figure with variables for the dimensions of a figure and an abstract method area(). Create two sub classes of this class: Rectangle and Triangle. Override the method area() to calculate area for each of them. Create objects of these two classes and prints the area. [05]
- (b) **Attempt Following.** [09]
1. Describe the use of super keyword with example.
 2. Explain how final modifier can be affect declaration of class, method and data?
 3. What is constructor? Differentiate between constructor and method of class.

OR

Q - 2 Answer the following.

- (a) Write a java program to read one file name and one character as a command line argument. Then counts the number of occurrences of given character within that file. [05]
- (b) **Attempt Following** [09]
1. Describe the use of this keyword with example.
 2. Explain why java is platform independent?
 3. Explain the two forms of the import statement.

Q - 3 Attempt following.

- (a) Differentiate applet paint() and repaint() method. [02]
- (b) What is Thread? Draw and explain Life Cycle of Thread. [05]
- (c) Write an applet program which performs following. [07]

When an applet is started, it has one button named Start on it. When user clicks the button the text of the button is changed from Start to Stop. When user clicks again the text should change from Stop to Start. Moreover, when Start is clicked the applet background should be RED and when Stop is clicked the applet background should be WHITE.

OR

Q - 3 Attempt following.

- (a) What is Wrapper class? List out all available Wrapper class. [02]
- (b) What is exception? Explain five keywords which are used to generate and handle exception in java. [05]
- (c) Write a java program which has Menu with options Shape and Color. Shape has submenu like Circle, Rectangle and Line. Color has submenu like Blue, Red and Green. Depending on selection different shapes with different color should be drawn inside applet area. [07]

SECTION – II

Q - 4 Attempt following.

[07]

- (a) When the String objects are compared with == the result is true, if the strings contains the same values.
1. true 2. false 3. can not define 4. NaN
- (b) In the code bellow, what data type the variable x can have?
byte b1 = 5;
byte b2 = 20;
x = b1 * b2;
1. byte 2. short 3. boolean 4. int
- (c) When we implement Runnable interface we have implement the method
1. start() 2. run() 3. runnable() 4. main()
- (d) Which of the following statements are valid array declarations?
1. int number (); 2. float average[]; 3. double[] marks; 4. number int[];
- (e) What is the error in following code?
class Test
{
 abstract void function ();
}
1. No error 2. method should be declared as static
3. class should be declared as abstract 4. class should be declared as public
- (f) What is the output of the following code?
int x = 33;
x = x >> 2;
1. 16 2. 32 3. 8 4. 4
- (g) class Base
 { int x = 10; }


```

class Derived extends Base
{ int x = 20; }
public static void main(String a[])
{
    Base b = new Base ();
    Derived d = new Derived ();
    Base bd = new Derived ();
    System.out.println( b.x + "," + d.x + "," + bd.x );
}

```

Will produce the output.

1. 10,20,20 2. 10,20,10 3. 20,10,20 4. 20,20,10

Q - 5 Attempt following.

- (a) A customer registers as a member by filling form and mailing it back to the company. No transactions for a period of one-year means he is removed from list and needs to reapply. A member informs the company for any change in personal details. A member makes an order by filling out a sales order form and then faxing it to the company (alternative is by phone). The assistant always checks the validity of the membership. Order processing clerk checks the availability of the ordered items and hold them for the order. If all items are available, clerk schedules delivery. Inventory control clerk controls and maintains the level of stock and responsible for reordering. For queries member phone the assistant. Members may return the defective goods within 30 days and get their money back. Each task carried out by the system has the name and ID of the staff concerned recorded in to the system. Draw Activity Diagrams. [05]
- (b) Define Data Flow Diagram, Data Flow and Control Flow. Draw Data Flow Diagram for money transaction process of ATM system [09]

OR

Q - 5 Attempt following.

- (a) Draw the Use-case Diagram for conducting Entrance Exam and Admission Process for Engineering Students in Gujarat. [05]
- (b) A bank has a computerized customer accounting/transaction system. The system can be accessed by ATMs which communicate with the central computer which clears transactions. Each ATM accepts a pass card interacts with the user, communicates with the central system for clearance purposes, dispenses cash, processes various error messages and allows access to additional services include paying bills, printing of statement, credit transfer. Users may only use their card to the maximum value of Rs.200.00 and not more then five times in one day. [09]
1. Identify the main classes their associations and attributes.
 2. Annotate the model with relevant constraints.
 3. Suggest how might generalization/specialization be introduced to your model?

Q - 6 Attempt following.

- (a) **Attempt any TWO** [08]
1. Explain the notation for the State Diagram.
 2. Explain: Multiplicity, Role name, Qualifier, Ordering with example.
 3. Describe Object Oriented Methodology in details.
- (b) **Attempt any THREE** [06]
- Differentiate between
1. Physical object and Conceptual object
 2. Object Design and System Design
 3. Aggregation and Association
 4. Class inheritance and Interface Inheritance
